Virtual University of Pakistan

BIO201 BIO202 BIO301 BIO302

MCQs for Midterms

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BIO201 MCQs Mids

Q

……energy is the energy of motion

(a) Kinetic

(b) Potential

(c) Chemical

(d) Atomic

Q

Ribosomes can be free or attached with

(a) golgi apparatus

(b) lysosomes

(c) endoplasmic reticulum

(d) necleus

Q

Which organ/molecule can convert light energy

(a) NAD

(b) Retina

(c) Biotin

(d) Coenzyme A

Q

Beta pleated sheet is found in

(a) Primary protein structure

(b) Secondary protein structure

(c) Tertiary protein structure

(d) Quaternary protein structure

Q

Inhibitor that binds to a site other than active site of enzyme

(a) Competitive inhibitor

(b) Irreversible inhibitor

(c) Reversible inhibitor

(d) Non competitive inhibitor

Q

lipid act as a following in which many components of cell membrane floats

(a) Pond

(b) Lake

(c) River

(d) Stream

Q

How many macromolecules a cell has....

(a) 2 (Maybe)

(b) 4

(c) 6

(d) 8

Q

Beta pleated sheet is found in

(a) Secondary protein structure

(b) Primary protein structure
(c) Tertiary protein structure
(d) Quaternary protein structure
Q
Cell membrane expels particles by vesicle formation
(a) pinocytosis
(b) Phagocytosis
(c) Endocytosis
(d) Exocytosis
Q
Molecule of phospholipids has ----------
(a) hydrophobic & Hydrophilic domain
(b) **hydrophobic only**
(c) hydrophilic only
(d) hydrogen linkage
Q
If one gene prevent the expression of other gene
(a) Pleiotropy
(b) Epistasis
(c) Synapse
(d) Tetrad
Q
Phenylalanine hydroxylase metabolizes
(a) Phenylalanine to Lysine
(b) Phenylalanine to Tyrosine
(c) Tyrosine to Phenylalanine
(d) Lysine to Phenylalanine
Q
Restriction enzymes inhibit the growth of
(a) Bacteriophage
(b) Bacteria
(c) Algae
(d) None
Q:
Missing of enzyme homogentisic acid oxidase caused
(a) Phenylketonuria
(b) Parkinson disease
(c) Alkaptonuria
(d) Cystic fibrosis
Q
Microtubules are originated from
(a) Cytoskeletal
(b) Nucleus
(c) Proteins
(d) Centrosomes
Q

The disintegration and condensation of nuclear material takes place in
(a) prophase II
(b) anaphase II
(c) metaphase II
(d) telophase II
Q
RNAs in large subunit of Eukaryotic ribosome
(a) 1
(b) 2
(c) 3
(d) 4
Q
Centromeres separate during whose anaphase:
(a) Mitosis
(b) meiosis I
(c) meiosis II
(d) None
Q
Enormous genetic information is present in
(a) RNA
(b) DNA.
(c) Nucleolus
(d) Nucleoplasm
Q
Okazaki fragments are joined by
(a) Ligase
(b) Gyrase
(c) Helicase
(d) Primase
Q
How much water present in a cell.....
(a) 50%
(b) 60%
(c) 70%
(d) 80%
Q
Charge on DNA is
(a) Positive
(b) Negative
(c) Neutral
(d) None

Q
Which of the following stabilizes cell shape
(a) DNA
(b) RNA

(c) **Actin**
(d) Myosin
Q
……energy is the energy of motion
(a) Kinetic
(b) Potential
(c) Chemical
(d) Atomic
Q
Space between double membrane structure of endoplasmic reticulum is called
(a) Vesicles
(b) Leumen
(c) Cavity
(d) **endoplasmic space**
Q
----------is one of the example of co-factor
(a) Copper
(b) Biotin
(c) Haeme
(d) All
Q
lipid act as a following in which many components of cell membrane floats
(a) Pond
(**b) Lake**
(c) River
(d) Stream
Q
DNA replicates during.....
(a) G0 phase.
(b) G1 phase
(**c) G2 phase**
(d) S phase
Q
Difference between the alpha 1-4 glycosidic and beta 1-4 glycosidic bonds is on the basis of :
(a) Specificity
(b) Stability
(c) neutral pH
(d) size and shape
Q
Example of variable expressivity
(a) Dementia
(b) Huntington disease
(c) Alzheimer disease.
(d) Pancreatic necrosis
Q
Following has a major role in down syndrome

(a) Maternal age
(b) Fraternal age
(c) Child age
(d) Relative age
Q
DNA replication obeys
(a) Dispersive model
(**b) Conservative model**
(c) Semi-conservative model
(d) Distractive model

Q
Missing of enzyme homogentisic acid oxidase caused
(a) Phenylketonuria
(b) Parkinson disease
(c) Alkaptonuria
(d) Cystic fibrosis
Q
The RNA is --------- while DNA is ----------
(a) double stranded, double stranded
(b) single stranded, double stranded
(c) double stranded, single stranded
(d) partially double stranded, double stranded
Q
Ribosome after protein formation arrives at
(a) Lysosome
(b) Rough endoplasmic reticulum
(c) Smooth endoplasmic reticulum
(d) Golgi bodies
Q
The nucleus membrane is continuous with
(a) golgi apparatus
(b) ribosomes
(c) endoplasmic reticulum
(d) nucleus
Q
Power generating organelle of cell is:
(a) golgi bodies
(b) nucleus
(c) mitochondria
(d) cell membrane
Q
The disintegration and condensation of nuclear material takes place in
(a) prophase II
(b) anaphase II
(c) metaphase II

(d) telophase II
Q
An observable feature in the subject of genetics is called
(a) Trait
(b) Character
(c) Phenotype
(d) All
Q
Ribosome after protein formation arrives at
(a) Lysosome
(**b) Rough endoplasmic reticulum**

202 MCQs Mids BIO

Q
Glucose residues in amylose are linked by _____.
α (1,4) linkage
β (1,4) linkage
α (1,6) linkage
None of these

Q
The number of carbon atoms in lysine is _____.
4
6
8
10

Q
If the carbonyl group is present at the end of the monosaccharide then it is called _____.
Acid anhydride
Aldose
Ketose
None of these

Q
The number of stereoisomers for a molecule containing only one chiral carbon is _____.
16
2
4
8

Q
Seminal fluid is rich in _____.
Fructose
Lactose

Xylose
Arabinose

Q
Which of the following glycosidic linkage exists between two glucose units of maltose?

alpha 1,4
beta 1,4
alpha 1,6
alpha1,4 and alpha 1,6

Q
Amino sugar is formed by the replacement of the hydroxyl group at ------ of the parent monosaccharide with amino group.
C-1
C-2
C-3
C-6

Q
Which one of the following is a structural homopolysaccharide?
Cellulose
Starch
Glycogen
All of these
Q
Which one of the following amino acids is not specified by three letter codon?
Serine
Proline
Selenocysteine
Methionine

Q:
Acid dissociation constants are designated as _____.

PKa
K
Pa
None of these

Q
The pI of glutamate is _____.

1.2
5.6
3.22

6.7

Q
Which of the following amino acid is negatively charged at physiological pH?
Aspartate
Lysine
Alanine
Proline

Q
Which of the following refers to particularly stable arrangements of amino acid residues giving rise to recurring structural patterns?
Primary structure
Secondary structure
Tertiary structure
Quaternary structure

Q
Keratin is present in _____.
Hair
Nail
Hoof
All of these

Q
Which of the following amino acid is involved in the hydrophobic interactions in the tertiary structure of protein?
Phenylalanine
Cysteine
Glycine
None of these

Q
The free energy of a protein molecule is influenced by _____.
Hydrophobicity
hydrogen bonds
electrostatic interactions
All of these

Q
Which of the following diseases is caused by protein misfolding?
Alzheimer's disease
Huntington's disease
Parkinson's disease
All of these

Q
Iron is a _____.

non-metal
metalloid
transition metal
halogen

Q
Myoglobin can bind _____ molecules of oxygen.

One
Two
Three
Four
Q
The partial pressure of oxygen (pO2) in lungs is about _____.
30 mmHg
50 mmHg
100 mmHg
200 mmHg

Q
In sickle cell anemia, Glutamate is replaced with _____.

Arginine
Alanine
Methionine
Valine

Q
Oxy hemoglobin dissociation curve describe the relationship of ----------
Available Oxygen
Amount of carbon dioxide carried by hemoglobin
Fibrinogen
Oxy-hemoglobin curve stability
Q
Which one of the following is made up of two molecules of glucose?
Maltose
Starch
Cellulose
Amylose

Q
Which bond exists between the carbonyl group and oxygen of a hydroxyl group in ring structure of glucose?

Ionic
Covalent
hydrophobic interaction
both ionic and covalent

Q
In Benedict test, the formation of cuprous oxide is indicated by the formation of _____.
blue ppt
silver mirror
red ppt
none of these

Q
Glucose 6 Phosphate is formed by the addition of phosphate group to C6 of glucose by _____ linkage.
Ether
Ester
Ionic
none of these

Q
E. coli convert lactose of milk to lactic acid by action of _____.
Beta galactosidase
Amyloglucosidase
Alpha amylase
Maltase

Q
At any pH above pI, gycine has a net _____ charge and will move towards the _____.
negative, anode
negative, cathode
positive, anode
positive, cathode

Q
Which of the following amino acids contain two COOH groups?
Alanine
Glutamate
Glycine
Valine

Q
Which of the following amino acid is positively charged at physiological pH?
Lysine
Aspartate
Tyrosine

Alanine

Q
Peptide bond is _____ linkage between two amino acids.
Ester
Amide
Ether
None of these

Q
Which of the following shows the greatest tendency to form α helices in most experimental model systems?
Glycine
Alanine
Proline
Serine

Q
Immunoglobulin G (IgG) consists of _____.
Two heavy chains and two light chains
Four heavy chains
Four light chains
None of these

Q
The free energy of a protein molecule is influenced by _____.
Hydrophobicity
hydrogen bonds
electrostatic interactions
All of these

Q
Proteins are denatured by _____.
Temperature
Ph
organic solvents
temperature, pH, organic solvents

Q
About _____ of amino acid residues in the myoglobin are found in alpha helices.
98%
78%
27%
10%

Q

Hemoglobin is found exclusively in _____.
white blood cells
red blood cells
platelets
platelets and white blood cells

Q
The partial pressure of oxygen (pO2) in lungs is about _____.
30 mmHg
50 mmHg
100 mmHg
200 mmHg

Q
In sickle cell anemia, Glutamate is replaced with _____.
Alanine
Methionine
Valine
Arginine

BIO301 MCQs Mid

Q
Genetics can be studied by
One way
Two ways
Three ways
Four ways

Q
Example of Mendilian genetics
Gene linkage
Pliotropy
Epigenetic
None

Q
In nature variation is
Temporary
Consistent
Widespread
All

Q
Condition in which an intermediate phenotype is developed from cross of parental generation.
Co dominance
Incomplete dominance
Both
Over dominance

Q
Condition in which one gene prevent the expression of other gene
Polymorphism
Pleiotropy
Epistasis
Polygon

Q
Bacterial chromosome is made up of
DNA
Histone
Both
RNA
Q
structure for chromosome stability
Telomere
Centrosome
Centromere
Chiasmata

Q
Broken ends of chromosomes are
Fluffy
Sticky
Transparent
Non sticky

Q
Centromere is unable to work due to mutation in
CDE-I
CDE-II
CDE-III
CDE-IV

Q
XO system of sex is present in
Fuitfly
Grasshoper

Snake
Frog

Q
prokaryotic cell divides by
Budding
Grafting
Mitosis
Binary fission

Q
Chromosomes become fit in the central line
Anaphase
Metaphase
Telophase
Prophase
Q
Haploid cells are produced by
Mitosis
Meiosis
Both
Parthenogenesis

Q
Tetrads align at center
Telophase I
Prophase I
Metaphase I
Anaphase I
Q
Which one is true
Sister chromatids are not identical in mitosis
Sister chromatids are identical in meiosis
Sister chromatids are not identical in meiosis
Non sister chromatids are identical in mitosis

Q
Genes are present on chromosomes first time said by
Greogor Mendel
T.H.Morgan
Walther Flemming
T.J.Robert

Q
Cross for determination of genotype of an individual

Cross over
Hybrid cross
Dihybrid cross
Test cross

Q
In ABO blood group system
Allele A is dominant over allele B
Allele B is dominant over allele A
Alleles A and B show co-dominance
Alleles A and B show incomplete dominance

Q
If two genes are more close then frequency of cross over will be
High
Low
High and low
Medium

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Q:
The hydrophilic backbones of alternating deoxyribose and phosphate groups are inside the double helix, facing the surrounding water?
a) True
b) False
c) False in certain cases
d) a and c
Q:
Chromosomes are made up of.....
a) Protein and DNA
b) DNA and RNA
c) RNA and protein
d) a & c
Q:
Chromosome theory of inheritance by Thomas Hunt Morgan was presented in
a) 1913
b) 1912
c) 1911
d) 1910
Q:
Nucleotide consists of …… major molecules
a) 2
b) 3
c) 4
d) 5

Q:
How many amino acids are there in one compelete tern of DNA...
a) 6.3
b) 3.6
c) 36
d) 4.6
Q:
Nucleosome fold to form the _____ fiber
a) 20 nm
b) 30 nm
c) 40 nm
d) 50 nm
Q:
On which media, cells are preferred to be grown?
a) N14
b) Pb
c) Cs
d) Ra
Q:
DNA polymerase has _____ domains
a) 3
b) 4
c) 5
d) 6
Q:
An E.Coli chromosome is estimated to have about.....super-coiled loops
a) 500
b) 400
c) 300
d) 200
Q:
Separation of DNA is accomplished by _____.
a) Type I topoisomerase
b) Type II topoisomerase
c) Type III topoisomerase
d) DNA polymease
Q:
Which one of the following is not an ionizing radiation?
a) X-rays
b) Ultraviolet light
c) Gamma rays
d) β rays
Q:
Hydrolytic deamination of cytosine is estimated to take place about
a) 1000 to 5000 times a day in a mammalian cell
b) 100 to 500 times a day in a mammalian cell

c) 10 to 50 times a day in a mammalian cell
d) 1to 5times a day in a mammalian cell
Q:
What will happen to the DNA exposed to N-methylguanine
a) it will be oxidized
b) it will make dimers of the bases
c) it will lead to production of mono-adduct
d) it will form a basic site
Q:
Aflatoxins are produced by
e) Bacteria
f) Virus
g) Fungi
h) Parasites
Q:
.....are similar enough to the proper bases to get taken up by cells converted into nucleoside triphosphate and incorporated into DNA during replication.
a) Interclating agents
b) Base analogs
c) Amino acid analogs
d) None of these
Q:
Cells infected with phage .....provide the ideal system to find the true template for protein synthesis
a) T4
b) P4
c) S4
d) L4
Q:
RNA polymerase performs essentially the same reaction in the cells from bacteria to human that is the synthesis of .....
a) cDNA
b) DNA
c) RNA
d) DNA, RNA and Cdna
e) Q:
There are three human MutS homologs
a) MSH1, MSH2, MSH3
b) MSH2, MSH3, MSH4
c) MSH2, MSH3, MSH6
d) MSH2, MSH3, MSH5
Q:
In long patch repair, how many nucleotides are repaired...
a) 1
b) 4-6

c) 2-8
d) 10-14
Q:
Pyrimidine dimers are formed by...
UV radiation
Infra red
Ultrasound
nuclear radiation